

College Students Assessment

(Assignment, Quiz, and Practical Components)

This assessment is structured around the core subjects studied during the **Internship program**, including **Information Security Fundamentals, Cryptography, Steganography, Phishing Attacks, Social Engineering Techniques, and SQL Injection (with practical analysis of the intentionally vulnerable application testphp.vulnweb.com)**.

The assessment aims to systematically evaluate students' theoretical knowledge and practical understanding of **SQL Injection vulnerabilities**, encompassing foundational concepts, exploitation methodologies, and mitigation and prevention techniques, as presented in the referenced learning materials and established information security practices.

PART I: SQL INJECTION

1. Short Answer / Explanation Questions

Q1. What is SQL Injection (SQLi), and why does it pose a severe threat to web applications?

Q2. In the context of the referenced article, how does appending a single quote (') to a URL help identify a vulnerable SQL parameter?

Q3. Define and differentiate between the following types of SQL Injection attacks:

- Error-Based SQL Injection
- UNION-Based SQL Injection
- Blind SQL Injection (Boolean-Based and Time-Based)

Q4. What is the purpose of using the ORDER BY clause during SQL Injection reconnaissance, and how does it help an attacker determine the structure of an SQL query?



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Q5. Explain how SQL Injection can be used to extract database information such as the current user, database name, and database version using UNION SELECT. Provide an example query.

2. Practical / Hands-On Tasks

Task 1: Manual Vulnerability Testing

Instructions:

- Visit:
`http://testphp.vulnweb.com/listproducts.php?cat=1`
- Append a single quote (') to the cat parameter in the URL.
- Observe whether an SQL error message appears.

Deliverables:

- Screenshot of the SQL error response
- A short explanation of what this result confirms

Note:

This activity must be performed **only on the intentionally vulnerable test site provided**. Testing real or unauthorized websites is strictly prohibited.

Task 2: Enumerating Columns

Instructions:

- Using the vulnerable parameter identified in Task 1, test sequential ORDER BY values
(e.g., ORDER BY 1 --, ORDER BY 2 --, etc.).



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- Identify the highest value before the server returns an error.

Deliverables:

- A table showing each ORDER BY test and its result
- A conclusion stating the number of columns returned by the query

3. Multiple Choice Questions (MCQs)**Basic Level**

Q6. SQL Injection occurs when user input is:

- A. Stored correctly in the database
- B. Properly filtered by the application
- C. Not properly sanitized and concatenated directly into SQL queries
- D. Encrypted before use

Q7. Which SQL payload condition always evaluates to true and is commonly used in basic SQL Injection?

- A. 1=2
- B. ' 1 ' OR ' 1 ' = ' 1 '
- C. DROP TABLE
- D. ; SHUTDOWN

Q8. Blind SQL Injection is primarily used when:

- A. The database returns verbose error messages
- B. There is no visible output and success is inferred from application behavior
- C. The attacker wants to delete tables
- D. The server is offline

Q9. The main advantage of using parameterized queries in application code is to:

- A. Increase query performance
- B. Prevent SQL Injection by separating data from SQL logic
- C. Allow dynamic SQL creation
- D. Store more rows



- Q10.** Which SQL Injection technique combines results from two separate queries?
- A. Boolean-Based
 - B. Time-Based
 - C. UNION-Based
 - D. Error-Based
- Q11.** SQL Injection mainly targets which component of a web application?
- A. Front-end UI
 - B. Database layer
 - C. Web server hardware
 - D. Network firewall
- Q12.** Which symbol is most commonly used to comment out the remainder of an SQL query in MySQL?
- A. ##
 - B. /* */
 - C. --
 - D. //
- Q13.** Which input field is most commonly tested first for SQL Injection?
- A. Search box
 - B. Login form
 - C. URL parameters
 - D. File upload field
- Q14.** What happens when a vulnerable application directly concatenates user input into an SQL query?
- A. Faster execution
 - B. Secure authentication
 - C. Possibility of SQL Injection
 - D. Data encryption

Intermediate Level

- Q15.** Which SQL keyword is primarily used in UNION-based SQL Injection?
- A. JOIN



- B. INSERT
- C. UNION
- D. WHERE

Q16. The purpose of UNION SELECT NULL, NULL -- is to:

- A. Delete database records
- B. Identify the number of columns
- C. Update table values
- D. Encrypt query results

Q17. Blind SQL Injection is used when:

- A. Error messages are visible
- B. The database is offline
- C. No error or data output is shown
- D. The user has admin access

Q18. Which SQL Injection technique relies on response delay to confirm a condition?

- A. Error-Based
- B. Boolean-Based
- C. UNION-Based
- D. Time-Based

Advanced Level

Q19. Which SQL function is commonly used to introduce delay in time-based SQL Injection (MySQL)?

- A. WAIT()
- B. DELAY()
- C. SLEEP()
- D. TIMEOUT()

Q20. Which database structure stores metadata such as table names and column names?

- A. mysql.user
- B. sys.tables



- C. information_schema
- D. db.meta

Q21. What does the payload ' OR 1=1 -- attempt to achieve?

- A. Delete a table
- B. Bypass authentication
- C. Create a new user
- D. Shut down the database

Q22. Which of the following is NOT an effective SQL Injection prevention technique?

- A. Input validation
- B. Parameterized queries
- C. Stored procedures with parameters
- D. Displaying detailed database errors

Tool-Based & Security Concepts

Q23. Which automated tool is widely used for detecting and exploiting SQL Injection vulnerabilities?

- A. Nmap
- B. Metasploit
- C. sqlmap
- D. Burp Scanner

Q24. Web Application Firewalls (WAFs) help prevent SQL Injection by:

- A. Encrypting databases
- B. Blocking malicious request patterns
- C. Replacing the database server
- D. Disabling SQL queries

Q25. Ethical hacking labs such as testphp.vulnweb.com are mainly used for:

- A. Attacking real websites
- B. Practicing illegal hacking
- C. Learning and safely testing vulnerabilities
- D. Hosting production data



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PART II: CRYPTOGRAPHY

(VeraCrypt & OpenPGP)

Section A: Cryptography Basics

- Q1.** Cryptography is mainly used to ensure:
- A. Speed and performance
 - B. Confidentiality, Integrity, and Authentication
 - C. File compression
 - D. Network routing
- Q2.** Which cryptographic technique uses the same key for encryption and decryption?
- A. Asymmetric encryption
 - B. Hashing
 - C. Symmetric encryption
 - D. Digital signature
- Q3.** Which of the following is an example of asymmetric cryptography?
- A. AES
 - B. DES
 - C. RSA
 - D. Blowfish
- Q4.** The primary purpose of hashing is to:
- A. Encrypt data
 - B. Decrypt files
 - C. Verify data integrity
 - D. Compress data
-

Section B: VeraCrypt

- Q5.** VeraCrypt is primarily used for:
- A. Email encryption
 - B. Disk and file encryption



- C. Network monitoring
- D. Password cracking

Q6. VeraCrypt is a successor to which encryption software?

- A. BitLocker
- B. AxCrypt
- C. TrueCrypt
- D. GnuPG

Q7. Which encryption algorithms are supported by VeraCrypt?

- A. AES only
- B. RSA and ECC
- C. AES, Serpent, Twofish
- D. MD5 and SHA-1

Q8. What is the purpose of a hidden volume in VeraCrypt?

- A. Increase encryption speed
- B. Provide plausible deniability
- C. Backup encrypted files
- D. Share files publicly

Q9. VeraCrypt protects against brute-force attacks primarily by using:

- A. CAPTCHA
- B. Keyfiles only
- C. Increased key-derivation iterations
- D. Two-factor authentication

Q10. Which file system can be used inside a VeraCrypt volume?

- A. FAT, exFAT, NTFS
- B. ext4 only
- C. HFS only
- D. RAW only

Section C: OpenPGP / GnuPG

Q11. OpenPGP is mainly used for:

- A. Disk encryption



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- B. Secure email communication
- C. Database encryption
- D. File compression

Q12. Which tool is commonly used to implement OpenPGP?

- A. VeraCrypt
- B. BitLocker
- C. GnuPG (GPG)
- D. OpenSSL

Q13. OpenPGP uses which cryptographic approach?

- A. Only symmetric encryption
- B. Only hashing
- C. Hybrid encryption (symmetric + asymmetric)
- D. Plain text encoding

Q14. Which key is shared with others in OpenPGP?

- A. Private key
- B. Session key
- C. Public key
- D. Master password

Q15. What is the purpose of a digital signature in OpenPGP?

- A. Encrypt the message
- B. Verify sender authenticity and message integrity
- C. Compress data
- D. Hide metadata

Section D: Practical & Security Concepts

Q16. If the OpenPGP private key is lost, what is the consequence?

- A. Messages can still be decrypted
- B. Only new messages are affected
- C. Encrypted messages cannot be decrypted
- D. Public key can replace it



Q17. Which hashing algorithm is commonly used with OpenPGP?

- A. MD4
- B. SHA-256
- C. CRC32
- D. Base64

Q18. Keyfiles in VeraCrypt are used to:

- A. Replace passwords
- B. Store encrypted data
- C. Add additional authentication security
- D. Share volumes

Q19. Which scenario best suits VeraCrypt usage?

- A. Encrypting email communication
- B. Encrypting entire hard disks or USB drives
- C. Verifying software signatures
- D. Secure web browsing

Q20. Which scenario best suits OpenPGP usage?

- A. Encrypting operating systems
- B. Encrypting database servers
- C. Secure email and file exchange
- D. Password management

Section E: Advanced Concepts

Q21. VeraCrypt volumes remain secure even if stolen because:

- A. Data is hidden
- B. Strong encryption and authentication are used
- C. Files are compressed
- D. Internet connectivity is required

Q22. What is the “web of trust” in OpenPGP?

- A. Central certificate authority
- B. Password sharing mechanism
- C. Decentralized trust model
- D. Firewall rule set



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- Q23.** Which attack is mitigated by using digital signatures in OpenPGP?
- A. Brute-force attack
 - B. Man-in-the-middle attack
 - C. SQL Injection
 - D. Buffer overflow
- Q24.** Which encryption mode is commonly used with AES in VeraCrypt?
- A. ECB
 - B. CBC
 - C. XTS
 - D. OFB
- Q25.** Combining VeraCrypt and OpenPGP provides:
- A. Redundant encryption
 - B. Disk encryption and secure communication
 - C. Faster encryption speed
 - D. Malware protection
-

PART III: PHISHING ANALYSIS

Section A: Basics of Phishing

- Q1.** Phishing is best described as:
- A. Encrypting user data
 - B. A social engineering attack to steal sensitive information
 - C. A network scanning technique
 - D. A password recovery process
- Q2.** Which of the following is the most common target of phishing attacks?
- A. Servers
 - B. Routers
 - C. End users
 - D. Firewalls
- Q3.** Phishing attacks are primarily delivered through:
- A. Antivirus software
 - B. Emails and messages



- C. Operating system updates
- D. BIOS firmware

- Q4.** Which of the following is an example of sensitive information targeted by phishing?
- A. Browser version
 - B. Screen resolution
 - C. Passwords and OTPs
 - D. Public IP address
-

Section B: Types of Phishing

- Q5.** Phishing that targets a specific individual or organization is called:
- A. Whaling
 - B. Vishing
 - C. Spear phishing
 - D. Smishing
- Q6.** Voice-based phishing attacks are known as:
- A. Smishing
 - B. Pharming
 - C. Vishing
 - D. Spoofing
- Q7.** SMS-based phishing attacks are called:
- A. Vishing
 - B. Smishing
 - C. Whaling
 - D. Baiting
- Q8.** Phishing attacks that target high-level executives are called:
- A. Spear phishing
 - B. Pharming
 - C. Whaling
 - D. Clone phishing
-



Section C: Phishing Email Analysis

- Q9.** Which of the following is a common indicator of a phishing email?
- A. Correct grammar and spelling
 - B. Personalized official domain
 - C. Urgent or threatening language
 - D. Encrypted attachment
- Q10.** A mismatched URL (displayed text differs from actual link) indicates:
- A. Safe redirect
 - B. DNS error
 - C. Possible phishing attempt
 - D. CDN usage
- Q11.** Which email header field is most useful for tracing the sender's origin?
- A. Subject
 - B. To
 - C. Received
 - D. CC
- Q12.** What does a shortened URL in a phishing email usually attempt to do?
- A. Improve readability
 - B. Hide the malicious destination
 - C. Increase speed
 - D. Encrypt traffic
-

Section D: Technical Analysis

- Q13.** Which protocol helps prevent email spoofing?
- A. FTP
 - B. SPF
 - C. SNMP
 - D. DHCP
- Q14.** DKIM is used to:
- A. Encrypt emails
 - B. Verify email integrity and authenticity



- C. Block spam
- D. Compress attachments

Q15. Which tool is commonly used to analyze phishing URLs?

- A. Wireshark
- B. VirusTotal
- C. Metasploit
- D. Nessus

Q16. A phishing website often imitates:

- A. Gaming portals
- B. Banking and social media websites
- C. Educational websites
- D. News websites

Section E: Prevention & Awareness

Q17. The most effective defense against phishing is:

- A. Expensive hardware
- B. User awareness and training
- C. Faster internet
- D. High-end graphics cards

Q18. What action should be taken when a phishing email is received?

- A. Reply immediately
- B. Click the link to verify
- C. Report and delete the email
- D. Forward it to others

Q19. Multi-Factor Authentication (MFA) reduces phishing impact by:

- A. Encrypting emails
- B. Blocking websites
- C. Requiring additional verification beyond passwords
- D. Hiding URLs

Q20. Which browser feature helps protect users from phishing sites?

- A. Dark mode



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- B. Pop-up blocker
 - C. Safe browsing warnings
 - D. Download manager
-

Section F: Advanced / Scenario-Based

Q21. An email stating “Your account will be suspended in 24 hours” is an example of:

- A. Legitimate alert
- B. Fear-based phishing
- C. System notification
- D. Insider threat

Q22. Which phishing attack redirects users to a fake website even when the correct URL is entered?

- A. Phishing
- B. Spear phishing
- C. Pharming
- D. Smishing

Q23. Why are phishing emails often successful?

- A. Strong encryption
- B. Exploitation of human psychology
- C. Advanced malware
- D. High-speed servers

Q24. Which file attachment type is commonly used in phishing attacks?

- A. .txt
- B. .jpg
- C. .exe or .html
- D. .mp3

Q25. Email authentication protocols (SPF, DKIM, DMARC) mainly help prevent:

- A. Data loss
- B. Malware infection



- C. Email spoofing and phishing
 - D. DDoS attacks
-

Submission Guidelines

- Submit all written answers in a **single PDF document**
- Include **screenshots** for all practical tasks
- Use **tables, diagrams, or query examples** where appropriate
- **Cite references** if external sources are used beyond classroom or internship materials



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